LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	20
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Aircraft controls systems; mechanical, power assisted, fully powered & flies by wire type of controls surfaces.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Aircraft controls systems; mechanical, power assisted, fully powered & flies by wire type of controls surfaces.
5. Specific Objectives	23.1.5.1 Trainee will be able to explain the construction and functioning of aircraft flight controls systems and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of control surface diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary flight control surfaces?
(b) How does a pilot's input reach the control surface?
(c) What is the basic difference between mechanical and powered controls?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Mechanical control systems construction • Power assisted control systems operation • Fully powered control systems features • Fly-by-wire system architecture • Comparison of control system types	L & Ds	PPT, IB	15 min
(b)	• Detailed analysis of mechanical linkages • Hydraulic power assist mechanisms • Electronic signal transmission in FBW • Advantages and limitations of each type • Integration of control systems in modern aircraft	L & Ds	PPT, PDF	20 min
(c)	• Control surface actuation sequences • Failure modes in different control types • Maintenance considerations for each system • Future trends in control system design • Safety protocols for control system inspection	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a mechanical and a fly-by-wire control system.
(b) Describe the role of hydraulic power assist in aircraft controls.
(c) What are the key components of a fully powered control system?

12. Home Assignment and Follow Up	Prepare a brief note on the transition from mechanical to fly-by-wire systems, referencing the next lesson plan on mechanical remote controls.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Mechanical remote controls system: Teleflex, Bowden and use of control tube and cables	Refer to the master syllabus reference list	LP 21

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min